***Neuroanatomy*: Cerebellum & Diencephalon Gross Structure**

1. The gray matter of the cerebellum is located primarily on the surface as the cortex, but embedded deep within the white matter are how many pairs of deep cerebellar nuclei?
a. Two
b. Three
c. Five
d. Six
e. Four
2. Which of the deep cerebellar nuclei is the largest, most lateral, and exhibits a highly altered, crumpled-bag shape on gross cross-section?
a. Nucleus globosus
b. Nucleus emboliformis
c. Nucleus fastigii
d. Nucleus interpositus
e. Nucleus dentatus
3. Moving from lateral to medial across the white matter core of the cerebellar hemisphere, what is the correct spatial sequence of the deep cerebellar nuclei?
a. Fastigii, globosus, emboliformis, dentatus
b. Emboliformis, globosus, fastigii, dentatus
c. Dentatus, fastigii, globosus, emboliformis
d. Dentatus, emboliformis, globosus, fastigii
e. Globosus, emboliformis, dentatus, fastigii
4. Which of the deep cerebellar nuclei occupies a midline position, situated closest to the midline in the roof of the fourth ventricle?
a. Nucleus dentatus
b. Nucleus emboliformis
c. Nucleus nucleus
d. Center median nucleus
e. Nucleus fastigii
5. Together, the nucleus globosus and the nucleus emboliformis are frequently grouped together in functional neuroanatomy and referred to as the:
a. Dentate complex
b. Fastigial tract
c. Interposed nucleus (Nucleus interpositus)
d. Pulvinar complex
e. Geniculate body
6. Macroscopically, the cerebellum is located in which anatomical compartment of the cranial cavity, situated posterior to the pons and medulla?
a. Anterior cranial fossa
b. Middle cranial fossa
c. Superior sagittal sulcus
d. Subdural sinus
e. Posterior cranial fossa
7. The cerebellum consists of two large lateral expansions called hemispheres, which are joined together in the midline by a narrow, worm-like strip of tissue called the:

a. Folia
b. Tonsil
c. Flocculus
d. Vermis
e. Nodulus

8. The surface of the cerebellar cortex is heavily folded into narrow, parallel transverse ridges. These individual macroscopic ridges are known as:

a. Gyri
b. Sulci
c. Fissures
d. Lobules
e. Folia

9. The deep, prominent macroscopic groove that separates the superior surface of the cerebellum from the inferior surface is the:

a. Primary fissure
b. Posterolateral fissure
c. Chiasmatic sulcus
d. Horizontal fissure
e. Hypothalamic sulcus

10. On the superior surface of the cerebellum, which specific structural fissure separates the small anterior lobe from the large posterior lobe?

a. Horizontal fissure
b. Posterolateral fissure
c. Secondary fissure
d. Primary fissure
e. Median fissure

11. Located on the inferior surface of the cerebellum, the oldest functional part of the organ consists of the nodule of the vermis and a pair of lateral lobes called the:

a. Tonsils
b. Pyramids
c. Uvulae
d. Flocculi
e. Lingulae

12. The rounded lobules of the cerebellar hemisphere located immediately lateral to the uvula of the vermis, which can herniate through the foramen magnum during high ICP, are the:

a. Flocculi
b. Nodules
c. Cerebellar tonsils
d. Folia
e. Alae

13. The cerebellum is structurally connected to the posterior aspect of the brainstem by three massive pairs of white matter fiber tracts called:

a. Internal capsules
b. Corona radiata
c. Stria terminalis columns
d. Fornix columns
e. Cerebellar peduncles

14. Which specific white matter peduncle connects the cerebellum directly to the midbrain and carries the bulk of the efferent output from the deep nuclei?
a. Inferior cerebellar peduncle
b. Middle cerebellar peduncle
c. Superior cerebellar peduncle
d. Anterior commissure
e. Corpus callosum
15. The largest of the three peduncles, composed almost entirely of afferent fibers passing from the pontine nuclei into the cerebellar hemisphere, is the:
a. Superior cerebellar peduncle
b. Inferior cerebellar peduncle
c. Stria medullaris
d. Internal capsule
e. Middle cerebellar peduncle
16. The diencephalon is a midline structural block of the brain that surrounds which specific segment of the ventricular system?
a. Lateral ventricle
b. Fourth ventricle
c. Cerebral aqueduct
d. Central canal
e. Third ventricle
17. Grossly, the diencephalon is bounded superiorly by the fornix and the roof of the ventricle. Its inferior boundary is formed by structures that make up the:
a. Corpus callosum
b. Basal ganglia
c. Floor of the third ventricle (optic chiasm, infundibulum, mammillary bodies)
d. Tectum of the midbrain
e. Cerebellar vermis
18. The diencephalon is symmetrically divided into two halves by the narrow cleft of the third ventricle. Each half is macroscopically divided into quadrants by the:
a. Central sulcus
b. Lateral sulcus
c. Hypothalamic sulcus
d. Posterior commissure
e. Calcarine sulcus
19. Macroscopically, the diencephalon is partitioned into four primary components. These are the: HMH
a. Telencephalon, midbrain, pons, medulla
b. Caudate, putamen, globus pallidus, amygdala
c. Epithalamus, thalamus, hypothalamus, subthalamus
d. Tectum, tegmentum, crus cerebri, substantia nigra
e. Vermis, tonsil, flocculus, nodule
20. The large, egg-shaped mass of gray matter that forms the major part of the diencephalon and serves as the main relay station for the cerebral cortex is the:
a. Thalamus
b. Hypothalamus
c. Subthalamus
d. Epithalamus

e. Amygdala

21. The two thalami face each other across the third ventricle and are frequently connected across the midline of the cavity by a band of gray matter called the:

a. Anterior commissure
b. Posterior commissure
c. Interthalamic adhesion (massa intermedia)
d. Optic chiasm
e. Habenular commissure

22. The internal structural framework of the thalamus white matter contains a vertical sheet of fibers that divides the thalamus into anterior, medial, and lateral masses. This sheet is the:

a. External medullary lamina
b. Internal medullary lamina
c. Stria terminalis
d. Fornix
e. Corpus callosum

23. The large, prominent swelling located at the posterior, expanded end of the thalamus that overhangs the superior colliculus of the midbrain is the:

a. Anterior nucleus
b. Medial geniculate body
c. Lateral geniculate body
d. Pulvinar
e. Centromedian nucleus

24. Tucked beneath the posterior end of the thalamus are two small oval swellings. The swelling that serves as the primary relay station for the visual pathway is the:

a. Medial geniculate body
b. Lateral geniculate body
c. Pulvinar
d. Subthalamic nucleus
e. Pineal gland

25. The small oval swelling belonging to the thalamus that handles the relay of the auditory pathway from the inferior colliculus to the temporal cortex is the:

a. Lateral geniculate body
b. Pulvinar
c. Medial geniculate body
d. Pineal gland
e. Habenular nucleus

26. The component of the diencephalon located inferior to the hypothalamic sulcus, forming the lower lateral wall and floor of the third ventricle, is the:

a. Epithalamus
b. Hypothalamus
c. Subthalamus
d. Thalamus
e. Metathalamus

27. On the ventral surface of the brain, the hypothalamus exhibits a midline stalk that connects it to the pituitary gland. This structural stalk is the:

a. Mammillary body
b. Optic tract

c. Infundibulum
d. Tuber cinereum
e. Posterior commissure HMH

28. A pair of small, round, pea-sized masses of gray matter sit side by side on the ventral surface of the hypothalamus, posterior to the infundibulum. These are the:

a. Optic chiasm columns
b. Lateral geniculate bodies
c. Habenular nuclei
d. Mammillary bodies
e. Pineal recesses

29. The epithalamus forms the posterior part of the roof of the diencephalon. Its most prominent macroscopic feature is a small midline endocrine gland called the:

a. Pituitary gland
b. Adrenal gland
c. Pineal gland (epiphysis cerebri)
d. Thyroid gland
e. Thymus gland

30. The subthalamus is situated inferior to the thalamus and lies between the thalamus and the tegmentum of the midbrain. A major biconvex mass of gray matter within this region is the:

a. Caudate nucleus
b. Subthalamic nucleus
c. Red nucleus
d. Putamen
e. Globus pallidus